Monte Carlo Simulations

The Monte Carlo method was developed by a loosely managed team of mathematicians, statisticians, computer scientists and nuclear scientists in the mid-forties in the USA. It was driven by trying to prove the usefulness of the newly created computer – ENIAC – to post-war problems. The team came up with a computer algorithm that would use older statistical sampling techniques to solve a nuclear problem – that was the problem of neutron diffusion in fissional material (Metropolis, 1987). The solution to this problem helped the Americans build thermonuclear devices. The name for the technique allegedly came from the main statistician (Stanislaw Ulam)'s uncle who was supposed to regularly borrow money so that he could go to Monte Carlo – for gambling it is assumed.

A classic example is based on the original use-case for the Monte Carlo method:

"Consider a spherical core of fissionable material surrounded by a shell of tamper material. Assume some initial distribution of neutrons in space and in velocity but ignore radiative and hydrodynamic effects. The idea is to now follow the development of a large number of individual neutron chains as a consequence of scattering, absorption, fission, and escape. At each stage a sequence of decisions has to be made based on statistical probabilities appropriate to the physical and geometric factors. The first two decisions occur at time $t = 0$, when a neutron is selected to have a certain velocity and a certain spatial position. The next decisions are the position of the first collision and the nature of that collision. If it is determined that a fission occurs, the number of emerging neutrons must be decided upon, and each of these neutrons is eventually followed in the same fashion as the first. If the collision is decreed to be a scattering, appropriate statistics are invoked to determine the new momentum of the neutron. When the neutron crosses a material boundary, the parameters and characteristics of the new medium are taken into account. Thus, a genealogical history of an individual neutron is developed. The process is repeated for other neutrons until a statistically valid picture is generated. " (Metropolis, 1987).

Decisions are made using the pseudo-random number functions of the underlying hardware. If the random number is denoted by x, which is a real number in the set (0,1) experimental evidence recommends a function f= - ln(x) that models the actual distribution of neutron activity. (ln is the natural log function).